

2) Draw the derivation tree corresponding to the derivation in example 5.1

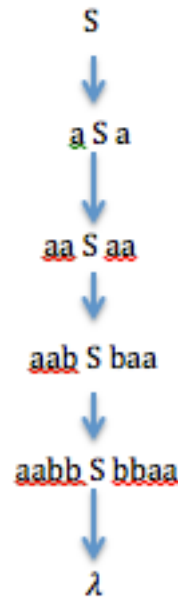
 $S \rightarrow aSa$ $S \rightarrow bSb$ $S \rightarrow \lambda$

Typical Derivation

 $S \Rightarrow aSa \Rightarrow aaSaa \Rightarrow aabSbaa \Rightarrow aabbSbbaa \Rightarrow aabbbaa$.

Derivation Tree

Grammar Tree

3) Give a derivation tree for $w = abbbaabbaba$ for the grammar in Example 5.2 $S \rightarrow abB$ $S \rightarrow aaBb$ $S \rightarrow bbAa$ $S \rightarrow \lambda$ **S**a b Bb b A aa a B bb b A a λ

5) Is the language in example 5.2 regular?

 $L(G) = \{ab(bbaa)^n bba(ba)^n \mid n \geq 0\}$ **Pumping Lemma,**

Proof:

Assume, for a contradiction, that L is regular so that there is a pumping length $m > 0$.Choose $w = ab(bbaa)^m bba(ba)^m$ so that $w \in L$ and $|w| \geq m$.Let $w = xyz$ be the decomposition of w given by the pumping lemma.

- ✓ $|xy| \leq m$.
- ✓ $|y| \geq 1$.
- ✓ $xy^i z \in L$ for all $i \geq 0$.

Choose $i = 2$ so that $w_1 = xy^2z$.

First, we can see that $bba(ba)^m$ is a suffix of w and so of w_1 , and it is the only string in L that has $bba(ba)^m$. Second, since $w_1 \neq w$ and w_1 has a suffix of $bba(ba)^m$, which is not in L , $w_1 \notin L$. These two conditions contradict the Pumping Lemma and hence our assumption was wrong, L is not regular.

7) Find context-free grammars for the following languages (with $n \geq 0$, $m \geq 0$)

$$c) L = \{a^n b^m : n \neq 2m\}$$

$$L = L_1 \cup L_2$$

$$L_1 = \{a^n b^m : n > 2m \geq 0\}, L_2 = \{a^n b^m : 0 \leq n < 2m\}$$

$$S \rightarrow S_1 \mid S_2$$

$$S_1 \rightarrow aaS_1b \mid A$$

$$A \rightarrow aA \mid a$$

$$S_2 \rightarrow aaS_2b \mid aB \mid B$$

$$B \rightarrow bB \mid b$$

It can be simplified as:

$$S \rightarrow S_1 \mid S_2$$

$$S_1 \rightarrow aaS_1b \mid aS_1 \mid a$$

$$S_2 \rightarrow aAS_2b \mid Ab$$

$$A \rightarrow a \mid \lambda$$

1) Find an s-grammar for $L(aaa^*b+b)$.

$$S \rightarrow aaAb \mid b$$

$$A \rightarrow Aa \mid \lambda$$

7) Construct an unambiguous grammar equivalent to the grammar in exercise 6.

$$S \rightarrow AB \mid aaB$$

$$A \rightarrow A \mid Aa$$

$$B \rightarrow b$$

$$S = \{aa^n b : b \geq 0\}$$

Unambiguous grammar:

$$S \rightarrow AB$$

$$A \rightarrow AX \mid a$$

$$B \rightarrow b$$

$$X \rightarrow a$$

11) Is it possible for a regular grammar to be ambiguous?

Yes, it is possible and it can be proved as follows.

We all know that λ is regular, thus a language that only generates the string λ is a regular language. Therefore, let's consider the following:

$$S \rightarrow S_1 \mid \lambda$$

$$S_1 \rightarrow \lambda$$

Since there are two leftmost derivations, S and S_1 , which generate the string λ this grammar is ambiguous. Hence this grammar is regular and ambiguous.

13) Show that the following grammar is ambiguous.

$$S \rightarrow aSbS \mid bSaS \mid \lambda$$

This can be shown as follows:

There are two leftmost derivations for the following string, abab

First Leftmost Derivation:

$$S \Rightarrow aSbS \Rightarrow abS \Rightarrow abaSbS \Rightarrow ababS \Rightarrow abab$$

Second Leftmost Derivation:

$$S \Rightarrow aSbS \Rightarrow abSaSbS \Rightarrow abaSbS \Rightarrow ababS \Rightarrow abab$$

Therefore we conclude that the grammar is ambiguous.